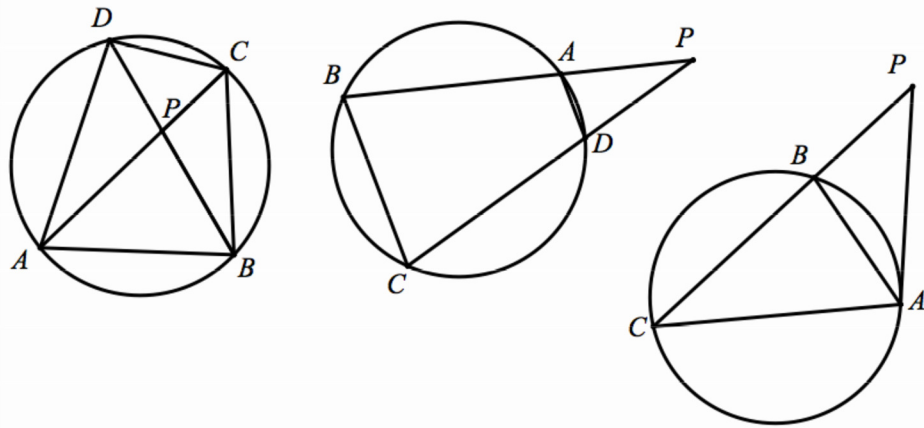


Olympiad Geometry – III

5. Power of a point to a circle

Intersecting chord theorem

- (a) Let $ABCD$ be a convex quadrilateral and let $P = AC \cap BD$. Then the points A, B, C, D are concyclic if and only if $PC \cdot PA = PB \cdot PD$.
- (b) Let $ABCD$ be a convex quadrilateral and let $P = AB \cap CD$. Then the points A, B, C, D are concyclic if and only if $PA \cdot PB = PC \cdot PD$.
- (c) Assume points P, B, C are collinear in this order and point A does not lie on this line. Then the line PA is tangent to the circumcircle of triangle ABC if and only if $PA^2 = PB \cdot PC$.



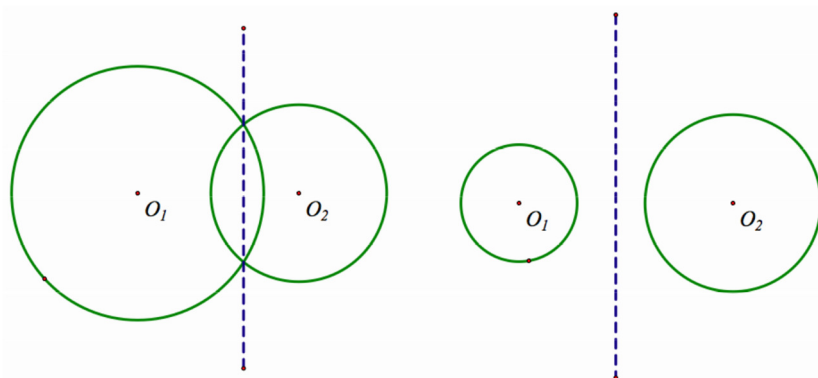
Power of a Point

Given point P and circle ω , let ℓ be an arbitrary line passing through P and intersecting ω at points A and B . Then the value of $PA \times PB$ does not depend on the choice of ℓ . Also, if P lies outside of ω and $PT, T \in \omega$, is a tangent to ω then $PA \times PB = PT^2$.

If we denote the centre of ω by O and its radius by R then $PA \times PB = |OP^2 - R^2|$.

The quantity $p(P, \omega) = OP^2 - R^2$ is called the power of point P with respect to circle ω .

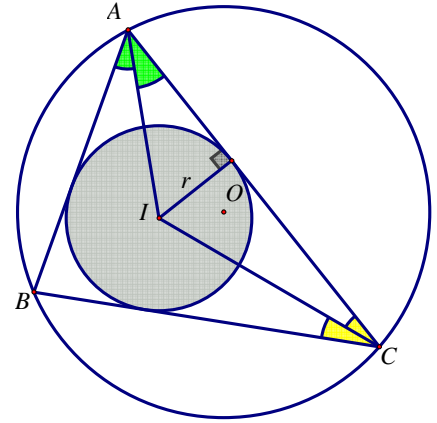
Given two circles ω_1 and ω_2 , the locus of points for which they have the same power of point with respect to the circles is a line called the radical axis of the two circles



In a rectangle coordinate system, the power of a point $P(a, b)$ to the circle $x^2 + y^2 + Dx + Ey + F = 0$ is given by $|a^2 + b^2 + Da + Eb + F|$.

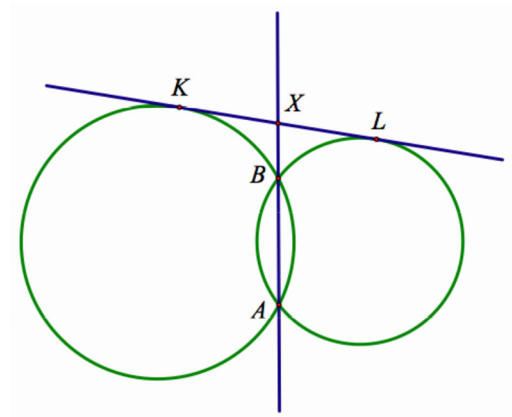
Euler's Identity

In $\triangle ABC$ with circumcentre O and incentre I , we have $OI^2 = R^2 - 2Rr$ and hence $R \geq 2r$ equality holds if and only if ABC is equilateral.

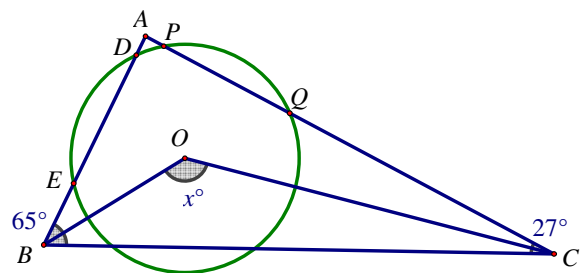
**Example 5.1 (Power of a point)**

Let the circles ω_1, ω_2 intersect at points A, B .

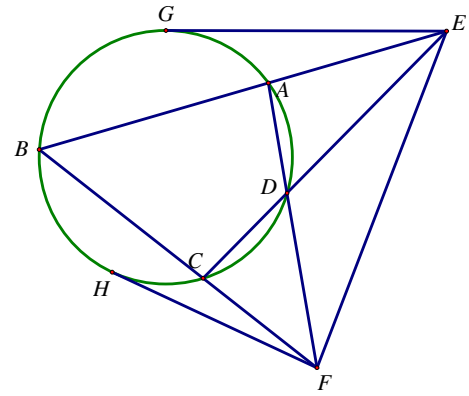
Denoted by K, L the points of tangency of the common external tangent with circles ω_1, ω_2 respectively. Then the line AB bisects the segment KL .

**Exercise:****1. (2019 PCIMC Heat Event (Senior) #20)**

In $\triangle ABC$, $\angle ABC = 65^\circ$ and $\angle ACB = 27^\circ$. A circle with centre O intersects AB at D and E and intersects AC at P and Q . If $AD \times AE = BD \times BE$, $AP \times AQ = CP \times CQ$ and $\angle BOC = x^\circ$, find the value of x .

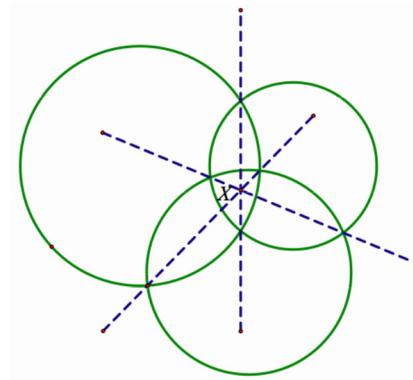


2. In the figure, EG and FH are tangents to the circle. Prove that $EF^2 = EG^2 + FH^2$



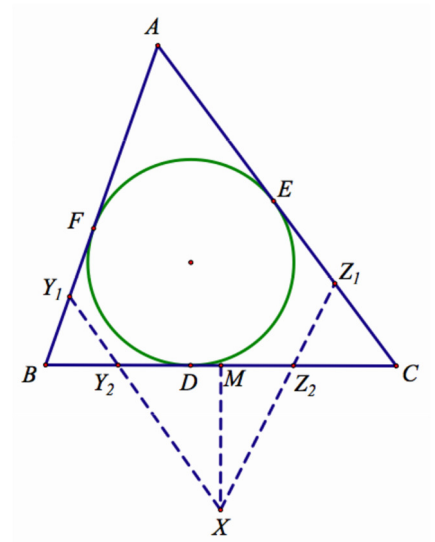
Radical Centre

Let $\omega_1, \omega_2, \omega_3$ be circles with pairwise distinct centres. Then their pairwise radical axes are either parallel or concurrent. The point of occurrence is called the radical centre of the three circles.



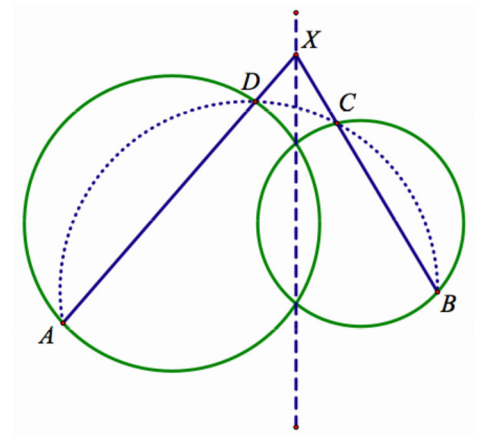
Example 5.2 (Radical Centre)

Let the incircle ω of triangle ABC touch BC, CA and AB at D, E and F respectively. Let Y_1, Y_2, Z_1, Z_2 and M be the midpoints of BF, BD, CE, CD and BC respectively. Let $Y_1Y_2 \cap Z_1Z_2 = X$. Prove that $MX \perp BC$.

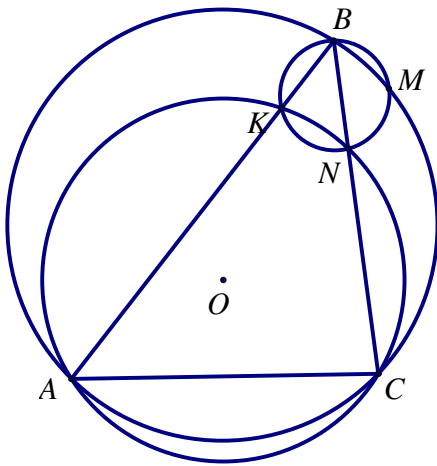


Radical Lemma**

Let the line L be radical axis of the circles ω_1, ω_2 . Let A, D be distinct points on ω_1 and let B, C be distinct points on ω_2 such that the lines AD and BC are not parallel. Then the lines AD and BC intersect at L if and only if $ABCD$ is cyclic.

**Example 5.3 (IMO 1985 P5)**

A circle with centre O passes through the vertices A and C of triangle ABC and intersects the segments AB and BC again at distinct points K and N , respectively. The circumscribed circles of the triangles ABC and KBN intersect at exactly two distinct points B and M . Prove that angle OMB is a right angle.



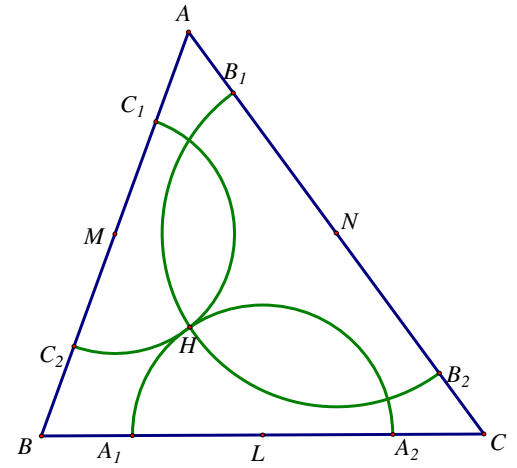
Example 5.4 (IMO 1995 P1)

Let A, B, C, D be four distinct points on a line, in that order. The circles with diameters AC and BD intersect at X and Y . Let P be a point on the line XY such that $P \notin BC$. The line CP intersects the circle with diameter AC at C and M , and the line BP intersects the circle with diameter BD at B and N . Prove that the lines AM, DN, XY are concurrent.



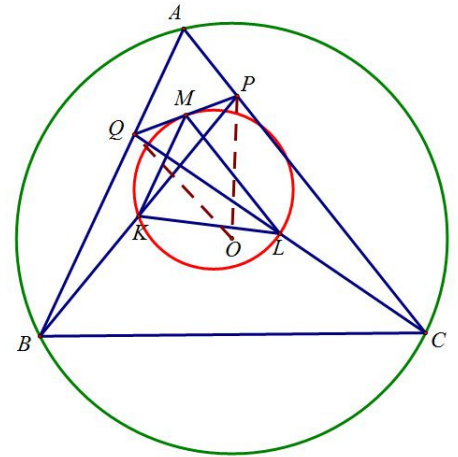
Exercise**Problem 1 (IMO 2008 #1)**

An acute-angled triangle ABC has orthocenter H . The circle passing through H with centre the midpoint of BC intersects the line BC at A_1 and A_2 . Similarly, the circle passing through H with centre the midpoint of CA intersects the line CA at B_1 and B_2 , and the circle passing through H with centre the midpoint of AB intersect the line AB at C_1 and C_2 . Show that $A_1, A_2, B_1, B_2, C_1, C_2$ lie on a circle.



Problem 2 (IMO 2009 #2)

Let ABC be a triangle with circumcentre O . The points P and Q are interior points of the sides CA and AB , respectively. Let K , L and M be the midpoints of the segments BP , CQ and PQ , respectively, and let Γ be the circle passing through K , L and M . Suppose that the line PQ is tangent to the circle Γ . Prove that $OP = OQ$.



Problem 3 (CGMO 2020 #1)

As shown in the figure, let $ABCD$ be a quadrilateral with $AB = AD$, $CB = CD$ and $\angle ABC = 90^\circ$. Points E , F lie on segments AB , AD respectively. Points P , Q lie on segment EF (P lies between E and Q), such that $\frac{AE}{EP} = \frac{AF}{FQ}$. Points X , Y lie on segments CP , CQ , respectively, such that $BX \perp CP$, $DY \perp CQ$. Show that X , P , Q , Y are concyclic.

